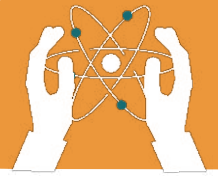


# UNIT 4

## FOOD CHAIN

### Learning objectives:

- Understand how energy flows in ecosystems through food chains.
- Identify the roles of producers, consumers, and decomposers in a food chain.
- Explain the importance of decomposers in breaking down organic matter and enriching the soil.



All living organisms present in this universe depend upon each other for food, shelter and protection. Food is the main source of energy which every organism needs to survive. The connection of "who eats whom" forms a **food chain**, which shows how energy is passed from one living thing to another. Let's explore how energy flows in ecosystem.

## Understanding Food Chains

A food chain refers to order of an event in ecosystem, which shows how is transferred from one to another. It usually starts with the **Sun**, which provides energy for plants to grow. This energy moves through different organisms, step by step.

## How Energy is Transferred?

**Sun:** It's the ultimate source of energy for all living organisms present in universe.

- **Producers:** Plants use sunlight to make their food through a process called photosynthesis and the organisms, which are able to produce the food by themselves, are known as producers.
- **Consumers:** Animals that eat plants or other animals for energy are considered as consumers.
- **Decomposers:** Tiny organisms that break down dead plants and animals, returning nutrients to the soil are known as decomposers.

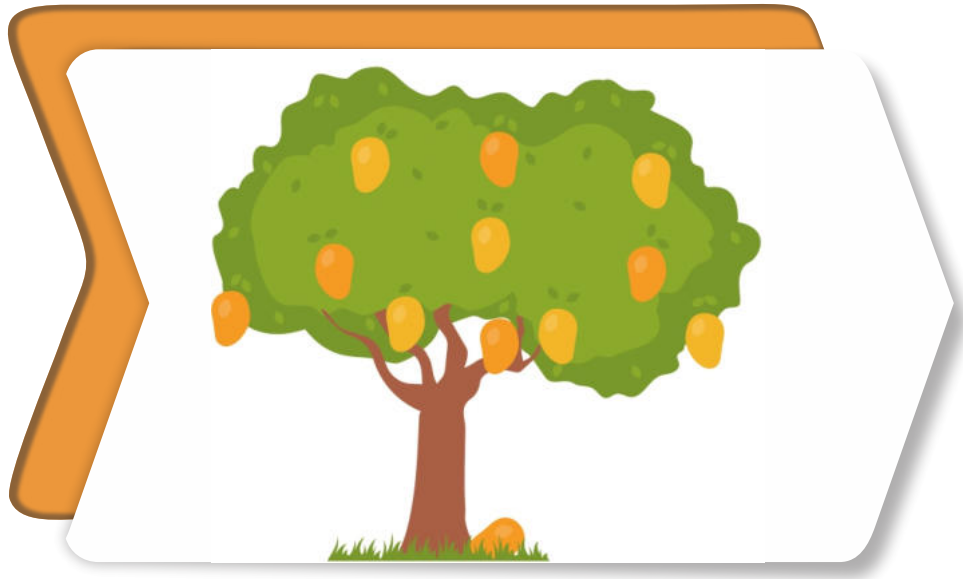
## Source of Energy in an Environment

### Producers

Plants are the producers.

They make their own food using sunlight, water, and carbon dioxide.

**Examples:** Wheat, grass, mango trees.



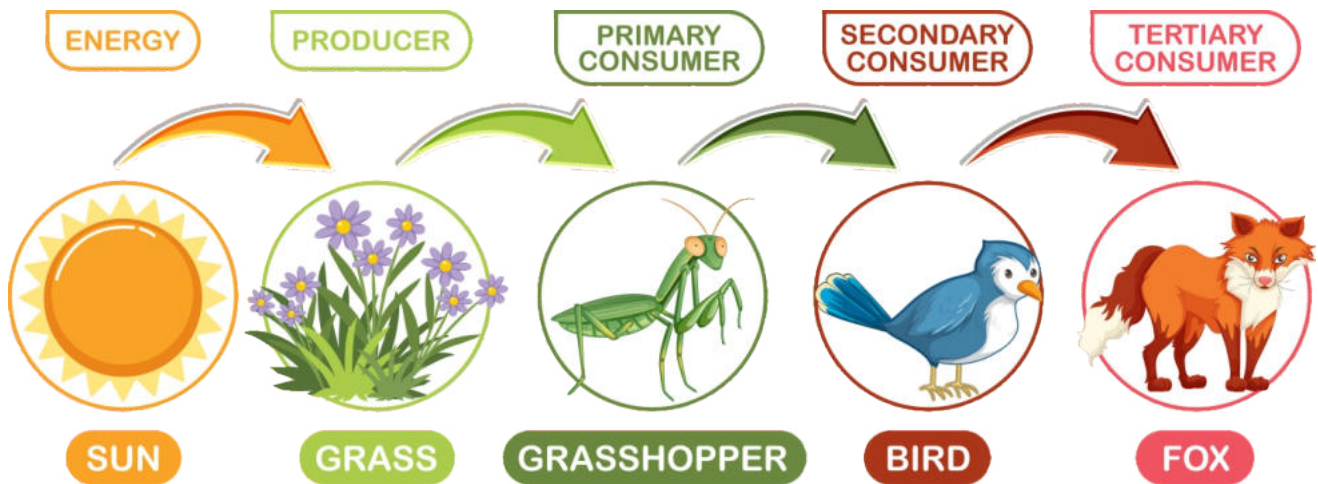
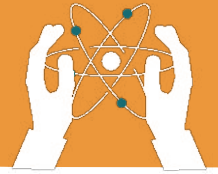
### Consumers

Consumers cannot make their own food. They depend upon producers as they eat plants or other animals.

There are three types of consumers:

- **Herbivores:** Animals that eat plants. (e.g., goats, cows)
- **Carnivores:** Animals that eat other animals. (e.g., lions)
- **Omnivores:** Animals that eat both plants and animals. (e.g., humans, bears, crows).

# FOOD CHAIN



## Decomposers

Decomposers are organisms like bacteria and fungi.

They break down dead plants and animals into nutrients, which enrich the soil.

Without decomposers, dead matter would pile up, and plants would not get nutrients.

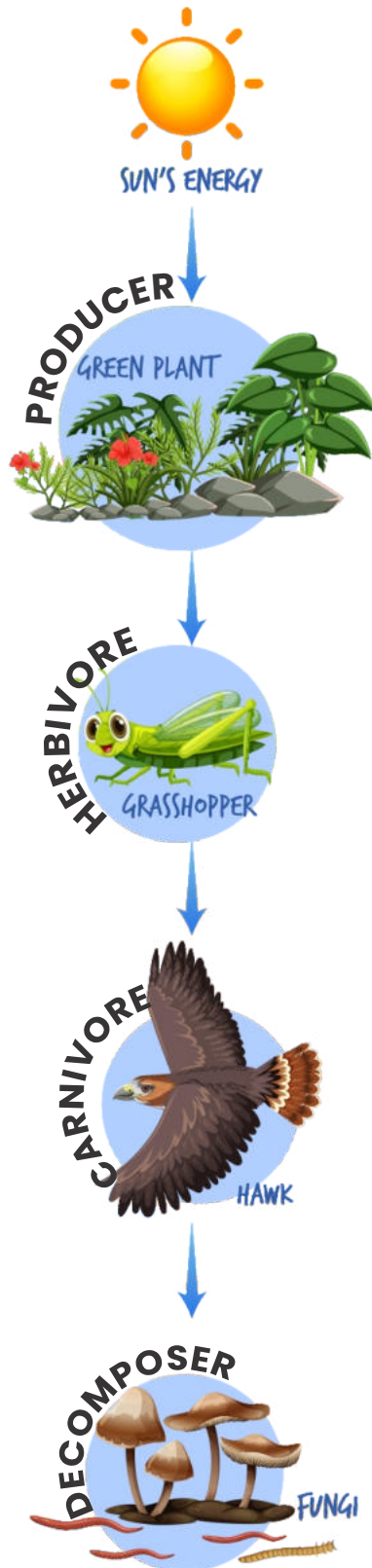


**Bracket fungus or shelf fungus**

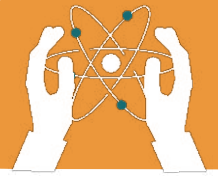


# FOOD CHAIN

## Example of a Food Chain

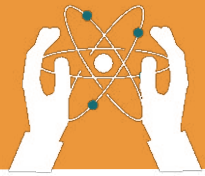


# FOOD CHAIN



## Fill in the blanks.

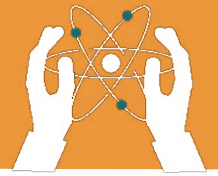
1. \_\_\_\_\_ is the prime source of energy.
2. \_\_\_\_\_ can make their own food.
3. Animals that eat only plants are called \_\_\_\_\_.
4. Lions and tigers are \_\_\_\_\_ because they eat other animals.
5. Decomposers break down \_\_\_\_\_ plants and animals.
6. \_\_\_\_\_ and fungi are examples of decomposers.
7. Humans are \_\_\_\_\_ because they eat both plants and animals.
8. Nutrients from decomposers enrich the \_\_\_\_\_.



## FOOD CHAIN

**Write “T” for True or “F” for False.**

1. Plants are producers because they depend on animals for food.
2. Carnivores eat both plants and animals.
3. The sun is the main source of energy for all food chains.
4. Decomposers return nutrients to the soil.
5. A goat is an example of a herbivore.
6. A food chain always ends with a carnivore.
7. Without decomposers, dead plants and animals would pile up.
8. Omnivores eat only plants.



## Choose the correct answer.

### 1. What is the starting point of all food chains?

- Grass
  - Sun
  - Animals
  - Soil
- 

### 2. Which of the following is a producer?

- Tiger
  - Goat
  - Wheat
  - Eagle
- 

### 3. Decomposers are responsible for?

- Eating plants
  - Returning nutrients to the soil
  - Hunting prey
  - None of the above
- 

### 4. Which is an example of a carnivore?

- Cow
- Lion
- Human
- Mushroom



**5. Animals that eat both plants and animals are called:**

- Carnivores
  - Omnivores
  - Herbivores
  - Producers
- 

**6. Grass, goat, and lion form a food chain. Which one is the producer?**

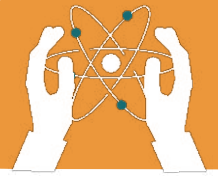
- Grass
  - Goat
  - Lion
  - None
- 

**7. What enriches the soil after plants and animals die?**

- Sunlight
  - Decomposers
  - Carnivores
  - Water
- 

**8. Humans are classified as:**

- Herbivores
  - Carnivores
  - Omnivores
  - Producers
-



## Answer the following questions.

- Q1. State the importance of food chain in ecosystem.**  
**Q2. Explain the role of producers in a food chain.**  
**Q3. Name three types of consumers and give one example of each.**  
**Q4. Why are decomposers important in the food chain?**  
**Q5. How does energy from the sun reach animals?**  
**Q6. Describe the difference between herbivores and omnivores.**  
**Q7. What happens to dead plants and animals in an ecosystem?**  
**Q8. Draw a food chain with examples to show transfer of energy.**



### Critical Thinking

- Why there are more number of herbivores than carnivores in an ecosystem?
- Think about the food you eat with its source.



## FOOD CHAIN



### Create Your Own Food Chain:



#### ACTIVITY

Using pictures or drawings, create a food chain starting from the sun. Include producers, herbivores, carnivores, and decomposers.

**Example:**

Sun → Grass → Rabbit → Fox → Fungi.